Presentation: Road Accidents in France

A MLOps Architecture for Road Safety Analytics

Objective: Building a MLOps framework to predict accident severity.
Focus: Integration of scheduling, experiment tracking, and containerizing for a multi-class classification model.

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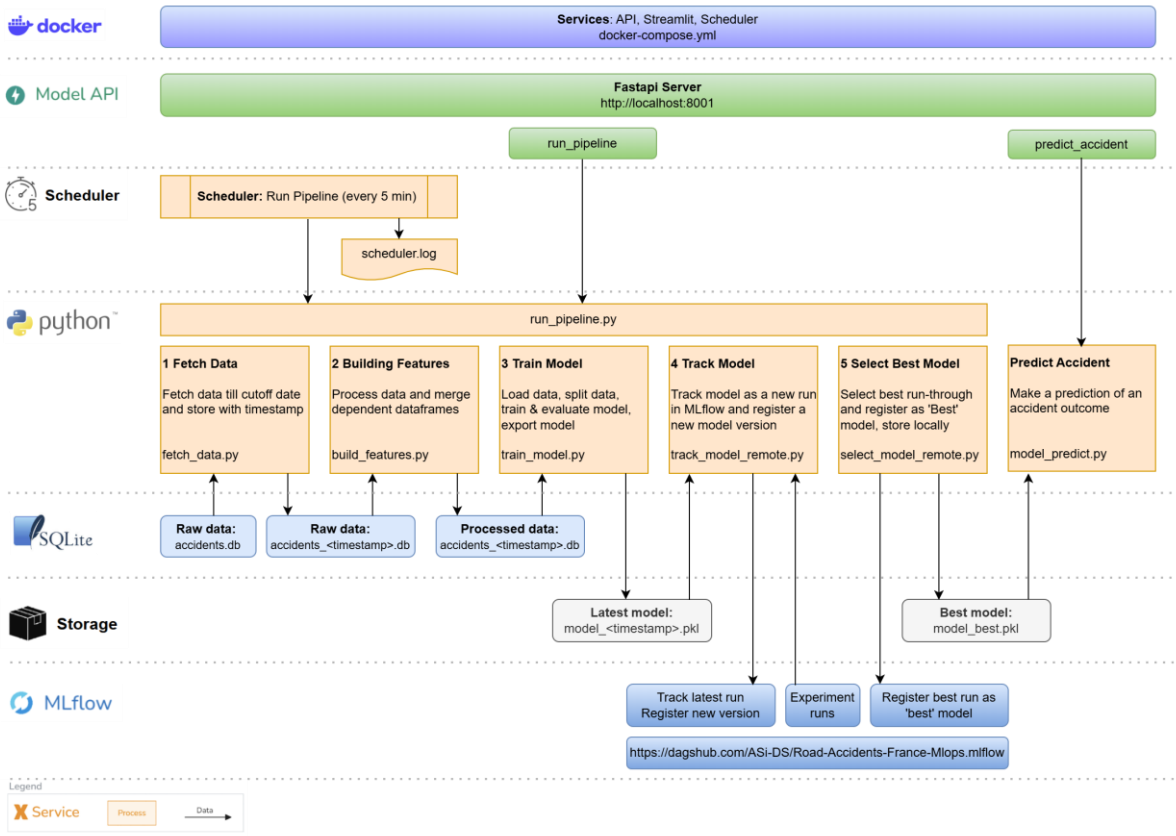
1. Architecture Diagram



2. Components

docker	**Docker**: Isolation and virtualization • **Standardization**: Guarantees a consistent environment across development, testing, and production. • **Encapsulation**: Packages the ML model, code, and specific library versions into a portable unit. • **Efficiency**: Provides process-level isolation that is faster and uses fewer resources than Virtual Machines.
FastAPI	**FastAPI**: Deployment in a microservice architecture • **Deployment**: Wraps service methods into a web API providing endpoints for clients to interact with the model. • **Decoupling**: Provides an API contract decoupling the service from clients. • **Security**: Provides authentication and authorization (not implemented).
mlflow DagsHub	**MLflow with DagsHub**: Remote experiment tracking & model versioning • **Experiment Tracking**: Logs parameters, metrics, and artifacts during training runs. • **Model Registry**: Serves as a central hub to version models. • **Access Control**: Provides built-in, team-based access controls. • **DagsHub Server**: Integrates Git and provides a hosted, remote MLflow tracking server offering a web-based UI to visualize MLflow experiment runs side-by-side.
Scheduler	**Scheduler**: Automated workflow • **Automation of Pipeline**: Executes ETL (Extract, Transform, Load), model training, tracking, and evaluation at specific intervals • **Dependency Management**: Ensures tasks happen in a defined order • **Error handling and logging**: Encapsulates error handling and logging

3. Scheduler and Pipeline

The pipeline is executed by a scheduler which runs every 5 min (or as indicated).

The scheduler serves to automate the workflow and makes sure that tasks are exectued in a defined order at specific intervals. Output and errors are logged to a dedicated scheduler.log

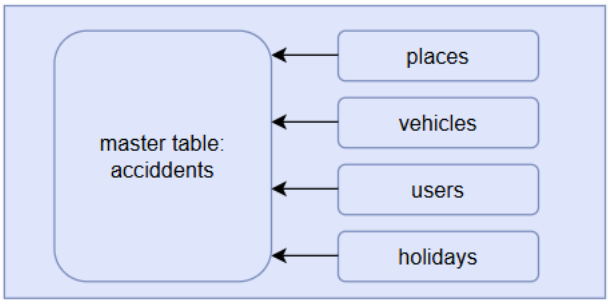
Alternatively, the pipeline can also be excuted via an API method: http://localhost:8001/run_pipeline

The pipeline consists of these steps:

4. Data loading and processing

Raw data consist of 5 dataframes, resp. tables:

- Accidents (master)
- Places
- Vehicles
- Users
- Holidays



Data loading and processing consists of several steps:

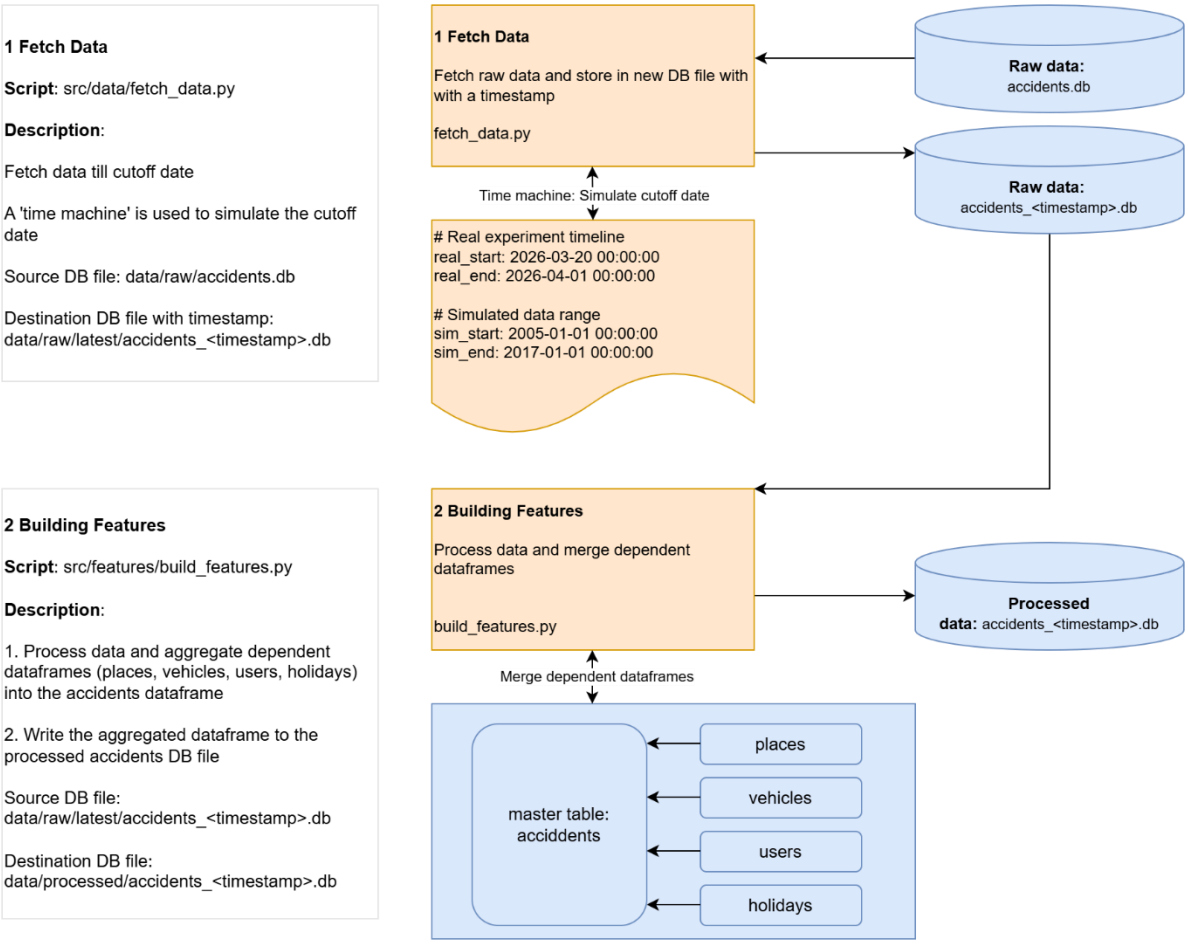
1 Fetch Data

- First, raw data are fetched from a raw DB file till a cut-off data:
  Since no new data are available, the full dataset (stored locally) is accessed till a cut-off date; the cut-off date is calculated using a time machine (s. below).
- The selected data are then written to a new DB file with the timestamp of the cut-off date:
  data/raw/accidents_<timestamp>.db

2 Build Features

- Data are being cleaned (eg. null rows are being dropped) and aggregated:
  Dependent dataframes (Places, Vehicles, Users, and Holidays) are merged into the Accidents dataframe
- The processed and aggregated data are then stored locally to another DB file:
- data/processed/accidents_<timestamp>.db

Data Processing Diagram:



Time Machine:

To simulate the cut-off date a ‘Time Machine’ is being used:

This concept works by proportionally mapping the real-world timeframe of the experiment to a simulated timeframe. It ensures the experiment progresses at the same relative pace, but within a different date range.

Logic:

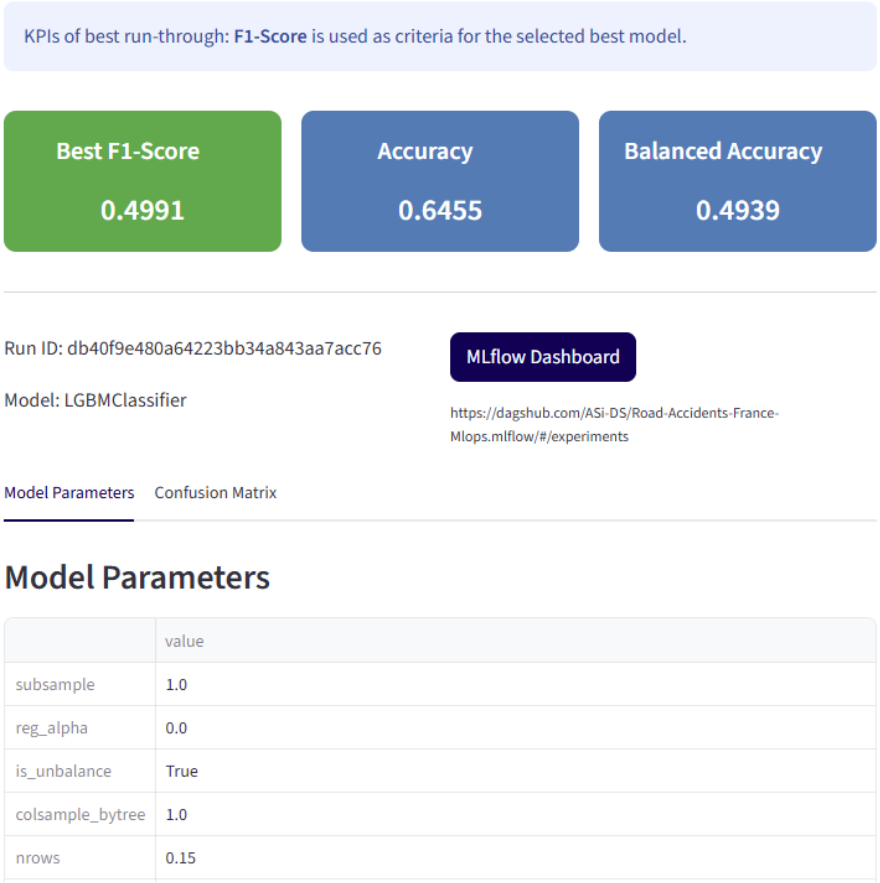
- **Normalization (Ratio):** Calculates progress of the real experiment duration (e.g. halfway between real_start and real_end corresponds to a ratio of 0.5).
- **Linear Mapping:** Applies the same percentage to the simulated duration. If the ratio is 0.5, it returns a timestamp halfway between sim_start and sim_end.
- **DB Filtering (Cut-off):** In the MLOps project, sim_time acts as a moving cut-off point. By querying the database for records where timestamp <= sim_time, "the present" is simulated in the past, hiding data that "hasn't happened yet" in the simulation.
- **In short:** It translates "How much time has passed since I started the script?" into "What date would it be in the historical dataset?"

5. MLflow with DagsHub: Remote experiment tracking & Model versioning

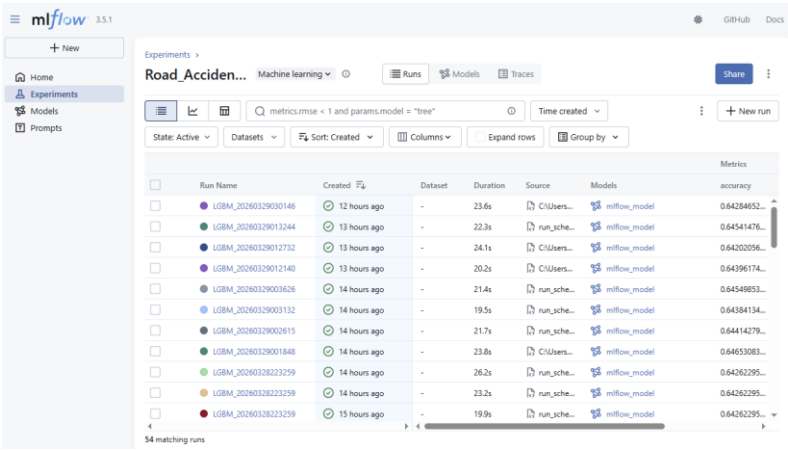
To track experiment run-throughs remotely, MLflow is used together with DagsHub.

The advantage of using DagsHub for remote tracking is the integration with Git, the storage of experiments and registered models remotely, and a web-based UI to visualize MLflow experiment runs.

Example of the MLflow dashboard in the Streamlit presentation:



Dagshub UI: https://dagshub.com/ASi-DS/Road-Accidents-France-Mlops.mlflow

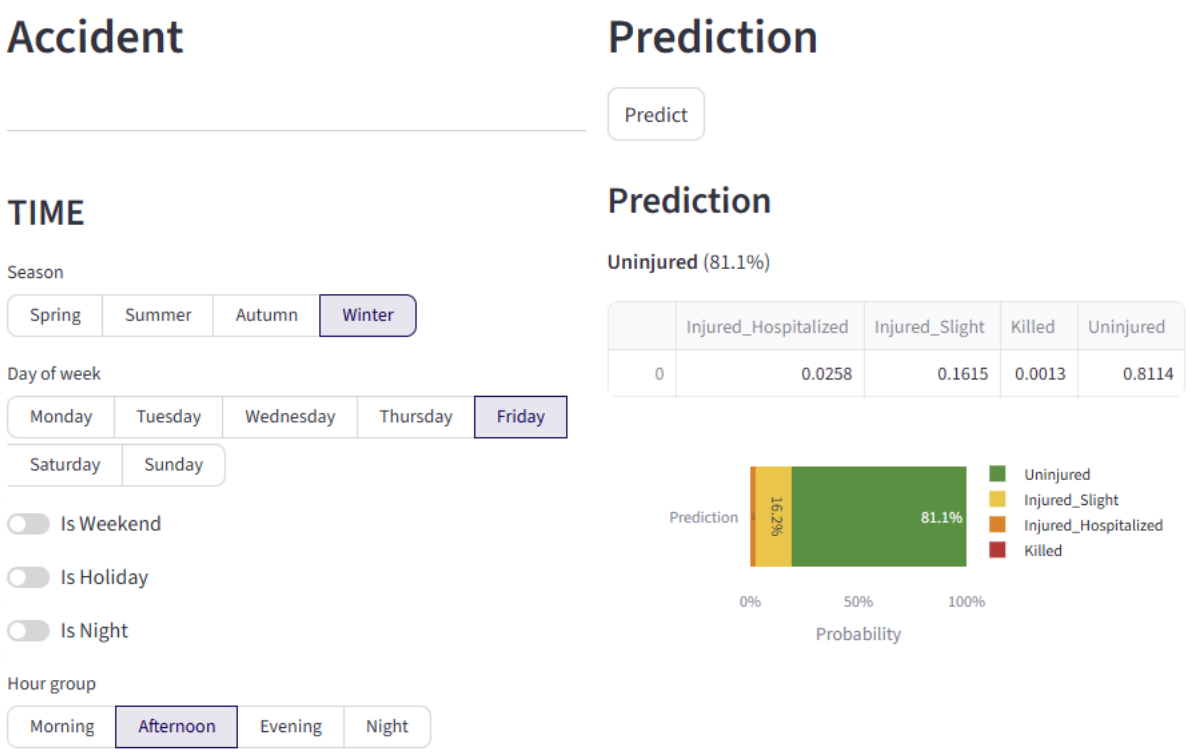


6. Accident prediction

The accident prediction uses the model to predict the outcome of an accident, based on features in the dataframe.

It can be called via an API method: http://localhost:8001/predict

Example of a prediction, Streamlit screenshot:



7. FastAPI Server Interface

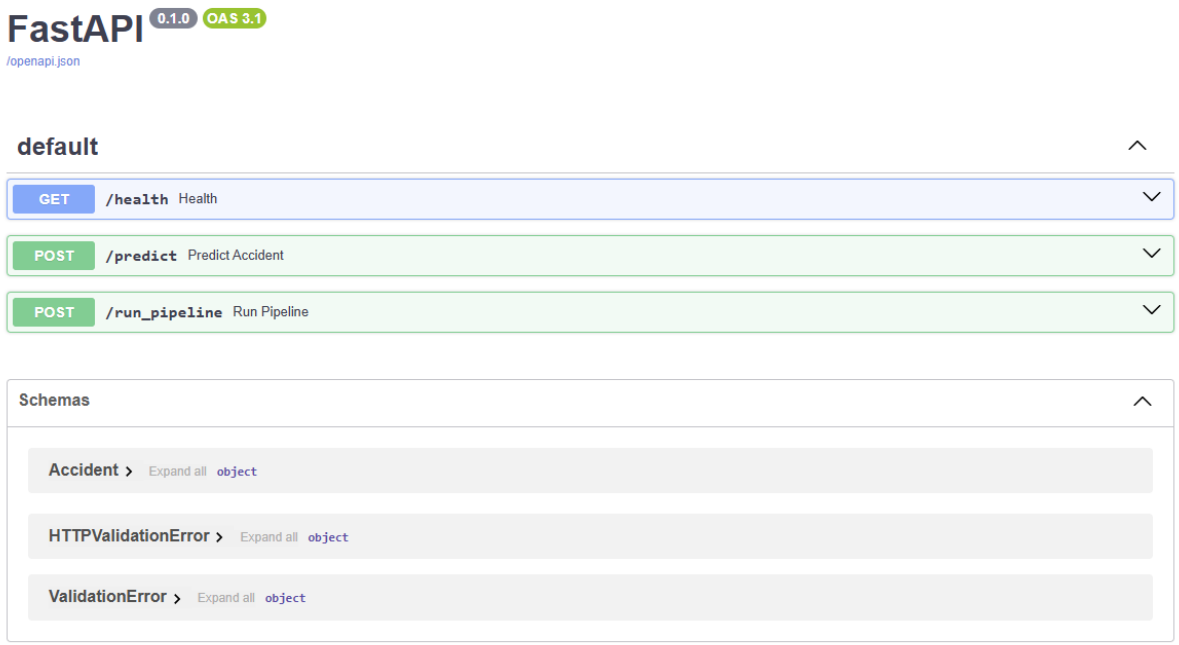
FastAPI provides a convenient deployment layer to implement a microservice architecture.

It wraps service methods into a web API and provides endpoints for clients to interact with the model. Thus, it decouples the service from clients by providing an API contract.

Local URL: http://localhost:8001/docs#/default

API methods currently deployed:

- Health
- Run_Pipeline
- Predict



8. Docker

Docker is a convenient method to isolate and virtualize an application. It creates consistency across different environments and packages the model, code and libraries into a portable unit.

These service components are currently included in docker-compose.yml

API	build:
	  context: .
	  dockerfile: Dockerfile.api
	ports:
	  - "8001:8001"
	volumes:
	  - ./models:/app/models
	environment:
	  MODEL_DIR: /app/models

Streamlit	build:
	  context: .
	  dockerfile: Dockerfile.streamlit
	ports:
	  - "8501:8501"
	environment:
	  API_BASE_URL: http://api:8001
	depends_on:
	  - api

Scheduler	build:
	  context: .
	  dockerfile: Dockerfile.api
	volumes:
	  - ./data:/app/data
	  - ./models:/app/models
	  - ./mlruns:/app/data/mlflow
	depends_on:
	  - api
	command: >
	  uv run python -m scripts.run_scheduler

9. Next Steps & Future Enhancements

Here are some suggestions for further improvements to this project:

1. Infrastructure & Security

- **Scheduling**: Replace the current scheduler script with an orchestrator like *Airflow*. This would allow for automatic retries and provide a visual interface to monitor pipeline health.
- **API Security**: Implement *OAuth2* or *API Key authentication* in FastAPI to ensure that only authorized services can trigger retraining or access the prediction engine.
- **Database Migration**: Transition from SQLite to a production-grade database like *PostgreSQL* for better concurrency and scalability.

2. Automated Testing & CI/CD Integration

- **Unit Testing for API Logic**: Implement *pytest* to automatically verify that endpoints handle data correctly.
- **Automated Testing in CI/CD**: Set up *GitHub Actions* to run a test suite automatically on every push to prevent "broken" code from reaching production.
- **CI/CD for Docker**: Automate the *Docker* build and deployment process using *GitHub Actions*. Every time the code changes, the API and Scheduler containers could be automatically built, tested, and updated.

3. Model Management (ML Life Cycle)

- **Dynamic Training Pipeline**: Shift from a "hard-coded" best model (based on the previous Data Science project) to a flexible training module.
- **Experiment Tracking**: Use *MLflow* to track and compare hyperparameter sweeps and different model architectures within the pipeline.